

RGAIG Clinical Questionnaire

Autism Spectrum Disorder (ASD) Assessment

Primary Data Collection Instrument — B2C (Caregiver) + B2B (Clinician) Tracks

Purpose: Two-track instrument capturing behavioural, psychological, social, and clinical data: the **B2C track** (Sections A–F) is filled by parents / caregivers; the **B2B track** (Sections G–I) is filled by clinicians / hospital admins. Both feed RGAIG mixed-methods analysis alongside EEG secondary data. Aligned with main.pdf B2C-primary, B2B-supporting framing.

Date: ____/____/____

Clinician: _____

Site: _____

Respondent: ☐ Parent ☐ Caregiver ☐ Teacher ☐ Self (if adult ASD)

Section A: Demographics (8 items)

#	Question	Response
A1	Patient's full name	_____
A2	Date of birth / Age	____/____/____ Age: ____
A3	Gender	☐ Male ☐ Female ☐ Other
A4	City / Country	_____
A5	Education level (patient)	☐ Pre-school ☐ Primary ☐ Secondary ☐ Adult
A6	Family history of ASD or neurodevelopmental disorder?	☐ Yes ☐ No If yes: _____
A7	Any prior ASD diagnosis?	☐ Yes (date: _____) ☐ No ☐ Suspected
A8	Current medications	_____

Section B: Behavioral Assessment (10 items)

[IV: Behavioral Severity]

Rate each behavior: 1=Never, 2=Rarely, 3=Sometimes, 4=Often, 5=Always

#	Behavioral Item	1	2	3	4	5	Clinical Construct
B1	Difficulty maintaining eye contact during conversation	☐	☐	☐	☐	☐	Social attention
B2	Engages in repetitive movements (hand flapping, rocking, spinning)	☐	☐	☐	☐	☐	Stereotypic behavior
B3	Resists changes in routine, environment, or schedule	☐	☐	☐	☐	☐	Rigidity / inflexibility
B4	Difficulty understanding others' emotions or facial expressions	☐	☐	☐	☐	☐	Theory of mind
B5	Unusual sensory responses (covering ears, avoiding textures, staring at lights)	☐	☐	☐	☐	☐	Sensory processing
B6	Difficulty with back-and-forth conversation or turn-taking	☐	☐	☐	☐	☐	Pragmatic language
B7	Intense focus on narrow interests (trains, numbers, maps)	☐	☐	☐	☐	☐	Restricted interests
B8	Difficulty transitioning between activities	☐	☐	☐	☐	☐	Executive function
B9	Echolalia (repeating words/phrases heard from others)	☐	☐	☐	☐	☐	Communication pattern
B10	Difficulty with pretend play or imaginative scenarios	☐	☐	☐	☐	☐	Symbolic play

Behavioral Score: _____ / 50 *Threshold: ≥ 30 = High concern; 20–29 = Moderate; < 20 = Low*

Section C: Psychological Assessment (10 items)

[IV: Psychological Profile]

Rate each item: 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly Agree

#	Psychological Item	1	2	3	4	5	Clinical Construct
C1	The individual experiences frequent anxiety or worry	☐	☐	☐	☐	☐	Anxiety
C2	The individual has meltdowns in overwhelming situations	☐	☐	☐	☐	☐	Emotional regulation
C3	Sleep quality has deteriorated in the past 6 months	☐	☐	☐	☐	☐	Sleep disruption
C4	The individual shows signs of depression or persistent sadness	☐	☐	☐	☐	☐	Depression
C5	The individual has difficulty coping with stress or frustration	☐	☐	☐	☐	☐	Coping mechanism
C6	Self-injurious behavior (head banging, biting, scratching) observed	☐	☐	☐	☐	☐	Self-harm risk
C7	The individual avoids new situations or environments	☐	☐	☐	☐	☐	Avoidance behavior
C8	Attention span is significantly shorter than peers	☐	☐	☐	☐	☐	ADHD comorbidity
C9	The individual becomes aggressive when routine is disrupted	☐	☐	☐	☐	☐	Aggression triggers
C10	The individual has food selectivity or eating difficulties	☐	☐	☐	☐	☐	Feeding behavior

Psychological Score: _____ / 50 *Threshold: ≥ 35 = Significant comorbidity; 20–34 = Moderate; < 20 = Mild*

Section D: Social Impact Assessment (8 items)

[IV: Social Functioning]

#	Social Item	1	2	3	4	5	Construct
D1	The individual faces bullying or social exclusion at school/work	☐	☐	☐	☐	☐	Social victimization
D2	Cannot participate in group activities independently	☐	☐	☐	☐	☐	Group functioning
D3	The condition has significantly affected family relationships	☐	☐	☐	☐	☐	Family impact
D4	Difficulty in educational or workplace performance	☐	☐	☐	☐	☐	Functional impairment
D5	Requires constant supervision or support	☐	☐	☐	☐	☐	Support needs
D6	Caregiver has experienced burnout, health issues, or job loss	☐	☐	☐	☐	☐	Caregiver burden
D7	Financial burden due to therapy, specialist visits, and support services	☐	☐	☐	☐	☐	Financial toxicity
D8	The individual has limited or no friendships	☐	☐	☐	☐	☐	Social isolation

Section E: Current Diagnostic Experience (7 items)

[DV: Diagnostic Delay]

#	Question	Response
E1	At what age did you first notice symptoms?	_____ months / _____ years
E2	At what age was formal diagnosis received?	_____ months / _____ years
E3	How many specialists were consulted before diagnosis?	_____ specialists
E4	Total cost spent on the diagnostic journey (approximate)	\$ _____ or INR _____
E5	Were you satisfied with the explanation of the diagnosis?	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
E6	Did you receive a clear treatment/intervention plan?	☐ Yes ☐ Partial ☐ No
E7	What was your biggest frustration during the diagnostic process?	_____

Section F: AI Screening Acceptance (5 items)

[DV: Adoption Intent]

Rate: 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly Agree

#	Statement	1	2	3	4	5
F1	I would use an AI-powered EEG screening tool for early ASD detection	☐	☐	☐	☐	☐
F2	I would trust an AI report that explains results with scientific evidence	☐	☐	☐	☐	☐
F3	I would prefer a 30-minute AI screening over a 2-year traditional pathway	☐	☐	☐	☐	☐
F4	I would use a wearable EEG device at home for continuous monitoring	☐	☐	☐	☐	☐
F5	I believe AI can complement (not replace) clinical judgment for ASD	☐	☐	☐	☐	☐

B2B TRACK — Clinician / Hospital Admin (Sections G, H, I)**Respondent:** ☐ Pediatric clinician ☐ Hospital admin ☐ DBA / IT lead ☐ Compliance officer**Section G: Governance Maturity (8 items)**

[IV: B2B Governance Maturity]

Rate each: 1=Not in place, 2=Ad-hoc, 3=Documented, 4=Measured, 5=Optimised

#	Governance Item	1	2	3	4	5	RGAIG Layer
G1	Our institution has documented AI policy covering fairness + accountability + transparency + privacy	☐	☐	☐	☐	☐	Governance
G2	We retain AI audit logs for ≥ 7 years per regulatory requirement	☐	☐	☐	☐	☐	Audit fabric
G3	We have a multi-jurisdiction compliance crosswalk (NIST + ISO + EU AI Act + DPDP)	☐	☐	☐	☐	☐	Governance
G4	We have an AI risk register reviewed by the board $\geq$ quarterly	☐	☐	☐	☐	☐	Governance
G5	We have a documented escalation path when AI outputs are uncertain	☐	☐	☐	☐	☐	HITL
G6	We score AI vendors against a governance-maturity instrument at procurement	☐	☐	☐	☐	☐	Procurement
G7	Our compliance team has signed off on cross-jurisdiction coverage	☐	☐	☐	☐	☐	Governance
G8	We monitor model drift + fairness gap continuously post-deployment	☐	☐	☐	☐	☐	MLOps

B2B Governance Maturity Score: _____ / 40 **Threshold:** ≥ 30 Optimised; 20–29 Managed; < 20 Initial

Section H: HITL Effectiveness (7 items)

[IV: HITL Moderator (H5)]

Rate each: 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly Agree

#	HITL Item	1	2	3	4	5	Tests
H1	The HITL gate adds $< 10\%$ overhead to clinician routine workflow	☐	☐	☐	☐	☐	H5
H2	Clinician override decisions are logged with reason codes + reviewed quarterly	☐	☐	☐	☐	☐	H5
H3	The AI tool surfaces dissent / council disagreement for clinician review	☐	☐	☐	☐	☐	H5
H4	I trust the AI output enough to act on routine cases under HITL gating	☐	☐	☐	☐	☐	H2, H5
H5	Clinician time-to-decision is shorter / unchanged with the AI tool vs without	☐	☐	☐	☐	☐	H4, H5
H6	The escalation path from caregiver to clinician is clear + auditable	☐	☐	☐	☐	☐	H5
H7	My institution would deploy this AI tool in a pilot for 6 months	☐	☐	☐	☐	☐	H5

Section I: Operational Integration + Procurement (6 items)**[DV: B2B Adoption]**

#	Integration Item	1	2	3	4	5	Tests
I1	The AI tool integrates with our EHR via standard APIs (FHIR / HL7)	☐	☐	☐	☐	☐	H4
I2	We can re-deploy a rolled-back model version within one operational shift	☐	☐	☐	☐	☐	H4
I3	Total cost of ownership (TCO) is within procurement budget tolerance	☐	☐	☐	☐	☐	H4
I4	The vendor provides a rollback + escalation plan documented in SLAs	☐	☐	☐	☐	☐	H5
I5	Cross-jurisdiction compliance evidence is current + auditable	☐	☐	☐	☐	☐	H3
I6	Our procurement decision will use the RGAIG 6-level maturity score	☐	☐	☐	☐	☐	H3

Variable Mapping: Independent & Dependent Variables (B2C + B2B Tracks)

Type	ID	Variable	Source (Section)	Scale	Hypothesis
IV	IV1	Behavioral severity score	Section B (B1–B10)	Ordinal (1–5)	H1, H3
IV	IV2	Psychological profile score	Section C (C1–C10)	Ordinal (1–5)	H2
IV	IV3	Social functioning score	Section D (D1–D8)	Ordinal (1–5)	H4
IV	IV4	Age at first symptom	Section E (E1)	Ratio (months)	H1
IV	IV5	EEG theta/beta ratio	Secondary data (EEG)	Ratio	H1, H3
IV	IV6	EEG frontal connectivity (PLV)	Secondary data (EEG)	Ratio	H1, H3
DV	DV1	ASD classification accuracy	ML output	Ratio (%)	H1
DV	DV2	Diagnostic delay (months)	Section E (E1 vs E2)	Ratio	H4
DV	DV3	Explanation satisfaction	Section E (E5)	Ordinal (1–5)	H2
DV	DV4	AI screening acceptance	Section F (F1–F5)	Ordinal (1–5)	H5
DV	DV5	Caregiver burden level	Section D (D6)	Ordinal (1–5)	H4
DV	DV6	Financial impact	Section E (E4)	Ratio (\$)	H4
MeV	MeV1	Trust in AI explanation	Section F (F2)	Ordinal (1–5)	H2→H5
MoV	MoV1	Family history of ASD	Section A (A6)	Nominal	H1
MoV	MoV2	Geographic location	Section A (A4)	Nominal	H4, H5

IV = Independent Variable; DV = Dependent Variable; MeV = Mediating Variable; MoV = Moderating Variable

Consent: I confirm that I have read and understood the purpose of this questionnaire and consent to the use of my/my child's data for research purposes in accordance with HIPAA/GDPR regulations.

Signature: _____

Date: ____/____/____